

SharP: Soft and Hard Prompt-Guided Augmentation with LLM for Low Resource Fake News Detection

Ziyu Luo^{a,b}, Wei Li^c, Haoping Huang^b, Kecheng Liu^d, Min Gao^{a,b,*}

^a*Key Laboratory of Dependable Service Computing in Cyber Physical Society (Chongqing University),
Ministry of Education, Chongqing 401331, China*

^b*School of Big Data and Software Engineering, Chongqing University, Chongqing 400044, China*

^c*Department of Plastic Surgery, Chongqing University Central Hospital, Chongqing Emergency Medical
Center, Chongqing 400000, China*

^d*Informatics Research Centre, University of Reading, Reading, United Kingdom*

Abstract

Fake news detection under low resource conditions is challenged by the scarcity of labeled data and the difficulty of capturing subtle deceptive patterns. In this work, we focus on *low resource* fake news detection, where only a small labeled set is available for training, while evaluation is performed under a fixed and unchanged test protocol. This setting is practically motivated by the high cost and ambiguity of expert fact-checking, as well as the rapid emergence of new topics and domains that are weakly covered by existing benchmarks, which together make large-scale, fully supervised training difficult to sustain. Existing data augmentation methods, such as synonym substitution, generative adversarial networks, and large language model (LLM)-based generation, often produce fluent but overly generic content, lacking the task-specific relevance needed for accurate detection. To address this, we propose **SharP**, a boundary-aware text generation framework guided by soft and hard prompts, which leverages LLMs as tools adapted to produce task-specific synthetic samples. SharP employs a dual prompt mechanism: hard prompts provide structured guidance and factual constraints, while soft prompts introduce learned semantic patterns from the data. These two types of prompts are combined to guide the generation process, enabling the model to produce samples that closely align with the characteristics of news. To further enhance detection performance, we incorporate a boundary-aware strategy that steers generation toward areas where the classifier is less confident, helping to clarify subtle distinctions between real and fake content. In addition, we adopt a dual objective optimization that balances semantic alignment with the source data and classifier feedback. This encourages the generation of samples that are both domain-consistent and helpful for refining decision boundaries, ultimately improving model robustness and generalization in low resource fake news detection.

Keywords: Fake News Detection, Large Language Models, Soft and Hard Prompting, Data Augmentation, Low Resource Learning

*Corresponding author.

Email addresses: ziyu.luo@stu.cqu.edu.cn (Ziyu Luo), liwei00128@cqu.edu.cn (Wei Li), haoping_huang@stu.cqu.edu.cn (Haoping Huang), k.liu@henley.ac.uk (Kecheng Liu), gaomin@cqu.edu.cn (Min Gao)

1. Introduction

The proliferation of fake news has emerged as a pressing global issue, driven by the unprecedented speed and reach of information dissemination enabled by social media platforms. Misinformation can severely distort public perception and trigger widespread consequences across domains such as politics, economics, and public health. Notable real-world examples include coordinated disinformation campaigns during elections and the circulation of false medical claims during the COVID-19 pandemic. Given its broad societal impact, fake news detection has become a focal research area across natural language processing, information retrieval, and computational journalism.

In recent years, a variety of techniques have been developed to tackle fake news detection, including textual content analysis, social propagation modeling, and multimodal fusion methods. However, the effectiveness of such models fundamentally relies on access to large-scale, high-quality labeled datasets. While existing benchmark datasets have facilitated significant progress, they remain limited in coverage, especially for emerging misinformation topics and early-stage detection tasks. Manual annotation of fake news is expensive, domain-specific, and subject to ambiguity, making the scalability of supervised learning approaches a persistent challenge. Consequently, enhancing model performance under low resource conditions has become a critical concern.

Data augmentation has emerged as a practical strategy to alleviate this bottleneck by expanding training sets with synthetic or transformed examples. Traditional text augmentation methods, such as synonym substitution, function word deletion, and back-translation, have shown effectiveness in improving model generalization (Kapusta et al., 2024). More recent efforts have introduced contrastive learning (Hua et al., 2023), multimodal synthesis, and GAN-based generation to further enrich the diversity of training data. With the advent of large-scale pre-trained language models, such as GPT and T5, researchers have begun to explore their potential for generating high-quality augmented samples in NLP tasks (Choi et al., 2025). These models offer strong semantic representation and generative capabilities, presenting a new frontier for data augmentation in fake news detection.

Nevertheless, existing LLM-based augmentation techniques often treat the generation process as a general-purpose paraphrasing task, lacking task-specific alignment and domain-level control. While these methods produce grammatically correct and fluent samples, they frequently fail to capture the deceptive stylistic patterns and subtle linguistic cues that distinguish fake news from real news. Moreover, most approaches do not consider the evolving weaknesses of the detection model during training. This results in a gap between the augmented data and the actual regions in the input space where the model is uncertain or error-prone.

Several recent studies have highlighted both the potential and the limitations of LLMs in this domain. Chen et al. (2025) empirically demonstrated that LLMs are effective at identifying textual misinformation but struggle to model propagation structures and social context. Raza et al. (2025) compared BERT-like encoders with autoregressive LLMs, finding that the former achieve better classification accuracy while the latter offer stronger robustness under noisy inputs. Liu et al. (2025) proposed a few-shot LLM-based framework that integrates external evidence and reasoning chains to enhance detection under ambiguity. These works collectively point to the need for more targeted LLM utilization strategies in fake news detection.

This paper addresses a central question: *How can LLMs be guided to generate augmented samples that are consistent with the domain, preserve the original labels, and*

effectively inform the classifier, thereby improving fake news detection in low resource settings? We argue that effective augmentation must satisfy three key requirements: maintaining semantic fidelity to preserve the true labels, ensuring domain specificity to capture deceptive traits, and being sensitive to the model’s uncertain predictions in order to improve classification performance.

To this end, we propose **SharP**, a boundary-aware text generation framework guided by soft and hard prompts. SharP transforms LLMs into task-aligned generators through prompt engineering, active boundary sampling, and dual objective optimization. The core idea is to control both *how* and *where* samples are generated: not only by instructing the model to preserve stylistic and semantic constraints, but also by strategically augmenting high-uncertainty regions in the classifier’s decision space.

Specifically, we develop a dual prompt architecture that combines structured *hard prompts*, which encode generation rules to enforce factual stance and stylistic fidelity, with *soft prompt* that are continuously optimized to inject domain-aware inductive bias into the LLM’s attention layers. This enables fine-grained control over the generation process without altering the model’s parameters. In addition, we introduce a boundary-sensitive dynamic augmentation strategy, which integrates an active learning loop to identify samples close to the classifier’s decision boundary. These samples are often associated with the uncertainty of the model and provide the most informative training signals. By focusing augmentation on these regions, SharP not only increases data diversity but also enhances model discrimination where it is most needed. Finally, we propose a dual objective optimization mechanism that jointly enforces feature-level semantic alignment and classification performance. The first objective ensures that augmented samples remain close to the originals in the feature space, maintaining semantic fidelity. The second uses the classifier’s error feedback on generated samples to iteratively refine the soft prompts, guiding the augmentation process to improve task relevance.

In this work, we define *low resource* fake news detection as a label-scarce training regime instantiated by restricting the amount of labeled training data. Concretely, we construct low resource settings by subsampling the labeled training set at multiple ratios, yielding progressively larger labeled pools while keeping the overall experimental setting consistent. All methods are trained under the same sampling ratios to enable a controlled comparison across different levels of supervision scarcity.

Taken together, SharP represents a principled integration of prompt-based control, uncertainty-driven sampling, and optimization-aware generation. By aligning augmentation with domain knowledge and classifier behavior, it substantially enhances model robustness and generalization in low resource fake news detection scenarios.

The contributions of this work are summarized as follows:

- A prompt-guided LLM-based augmentation framework was developed, in which structured hard prompts were integrated with learnable soft prompts to enable domain-consistent and label-preserving generation across diverse news topics. This framework was implemented and empirically validated on the PHEME dataset.
- A boundary-aware sample selection strategy was proposed, leveraging classifier uncertainty to dynamically guide augmentation toward informative regions in the decision space.
- A dual objective optimization scheme was designed to jointly promote semantic fidelity and discriminative enhancement, enabling end-to-end coordination between

the augmentation module and the detection model. The contribution of this scheme was validated via hyperparameter sensitivity analysis, confirming its role in enhancing model performance.

2. Related Work

2.1. Rumor Detection

Rumor detection on social media has evolved rapidly as misinformation becomes more prevalent and time-sensitive. An early milestone in neural rumor detection was to model textual signals for timely identification; for example, Gao et al. (2020) combined a character-level bidirectional language model with stacked LSTMs to capture sequential cues for early detection. A subsequent milestone was the shift from plain-text modeling toward leveraging richer supervision signals, incorporating heterogeneous user-generated context and multimodal evidence. For example, Qian et al. (2021) proposed a cross-modal detection framework that integrates textual content, social comments, and image-level sentiment information to improve robustness under diverse scenarios.

More recently, the literature has increasingly emphasized robustness under limited supervision, where labeled rumors are scarce and newly emerging events deviate from past distributions. To alleviate label scarcity, Zhao et al. (2021) introduced data augmentation for rumor detection to expand training signals, and Sun et al. (2022) further combined self-supervised learning with augmentation to improve representation stability. In parallel, feature fusion architectures have been advanced for platform-specific settings; Wang et al. (2022) studied Chinese social media and combined textual, visual, and user-level features via ERNIE and VGG-19 to strengthen multimodal integration.

Overall, the field has progressed from sequence-based textual modeling to multimodal and context-aware learning, and further toward augmentation- and self-supervision-driven robustness. Despite substantial progress, rumor detection still faces a persistent bottleneck in low resource and evolving-event scenarios, where annotation is costly and ambiguous and models must learn from limited, shifting evidence—motivating the need for task-aligned augmentation under scarce supervision.

2.2. Benchmark Datasets for Fake News and Rumor Detection

To situate our study on low resource fake news detection, we briefly review representative benchmarks that are widely used to evaluate models under controlled label budgets. A major line of work focuses on social-platform rumor verification, where the input often comes with conversational or event structures, such as the breaking-news threads in PHEME (Zubiaga et al., 2016) and the event-centric Twitter15/16 benchmarks (Ma et al., 2017a,b); complementary to English Twitter benchmarks, the Weibo rumor dataset provides a widely used Chinese microblog testbed, reflecting the linguistic and platform diversity of misinformation benchmarks (Ma et al., 2016). Another line targets claim/news-based fake news detection, including LIAR with fine-grained veracity labels for short fact-checked claims (Wang, 2017) and FakeNewsNet, which pairs news articles with associated social context from PolitiFact and GossipCop (Shu et al., 2020). In addition, CoAID offers a domain-specific benchmark for COVID-19 healthcare misinformation (Cui and Lee, 2020), enabling evaluation under rapidly evolving topics. Importantly, these datasets differ in annotation granularity, instance structure (e.g., claim-, article-, or thread-level), and available auxiliary signals; therefore, low resource settings are typically instantiated by subsampling the labeled pool while keeping the evaluation

protocol fixed. This makes them well-suited for assessing whether augmentation methods can improve generalization when labeled supervision is scarce, which is the primary setting addressed in our work.

Table 1: Representative benchmark datasets for fake news and rumor detection

Dataset	Source	Lang.	Scale	Labels / Structure
Social rumor verification				
PHEME (Zubiaga et al., 2016)	Twitter events	EN	330/4.8k	Conversation threads; rumor verification / rumor vs. non-rumor.
Twitter15/16 (Ma et al., 2017a,b)	Twitter events	EN	1490/818	Event-level rumor detection with reply structure.
Weibo (rumor) (Ma et al., 2016)	Sina Weibo	ZH	2,560 (1280 fake + 1280 true)	Post-level rumor vs. non-rumor (common benchmark variants).
News/claim fake news detection				
LIAR (Wang, 2017)	PolitiFact claims	EN	12.8k	6-way truthfulness; often binarized in detection.
FakeNewsNet (Shu et al., 2020)	PolitiFact/GossipCop	EN	1.1k/22.1k	Article-level fake/real with associated social context.
CoAID (Cui and Lee, 2020)	COVID-19 content	EN	1.9k (news articles)	Health misinformation benchmark (release-dependent labeling).

2.3. Data Augmentation for Fake News Detection

Data augmentation is a widely adopted strategy for improving text classification under limited labeled data, and it has been extensively explored in fake news detection. Early augmentation methods relied on syntactic or lexical perturbations. Back-Translation (Sennrich et al., 2016) generates semantically preserved paraphrases through round-trip translation, and EDA (Wei and Zou, 2019) further popularized lightweight operations such as synonym replacement, random insertion, swapping, and deletion. Subsequent work investigated interpolation-style augmentation at the representation level; MixText (Chen et al., 2020) extends the Mixup principle to textual inputs by mixing latent features to synthesize new training instances. In addition, Cheng et al. (2020) combined back-translation with adversarial training to promote diversity and robustness.

A major milestone in text augmentation is the emergence of large pre-trained language models, which shifts the paradigm from heuristic perturbations toward controllable, prompt-guided generation. LM-CPPF (Abaskohi et al., 2023) explored prompt-based few-shot paraphrasing with large models such as GPT-3 and OPT-175B, and Kesgin and Amasyali (2024) proposed leveraging the fill-mask capability within Transformer-based models for augmentation. Despite these advances, existing augmentation methods often produce fluent yet task-agnostic samples and underutilize domain-specific deceptive cues, underscoring the need for controllable, label-preserving, and domain-aligned generation in low resource fake news detection.

2.4. Soft Prompting in LLMs

Parameter-efficient prompting has emerged as an appealing alternative to full-model fine-tuning for adapting large language models. Early work demonstrated that optimizing continuous prompt representations can effectively steer model behavior with a small number of trainable parameters. A milestone in this line is Prefix-Tuning (Li and Liang, 2021), which learns task-specific continuous prefix vectors to condition generation while keeping the backbone frozen. Soon after, Prompt Tuning (Lester et al., 2021)

Table 2: Representative LLM-related studies for fake news detection and their reported results

Work	LLM role	Datasets / key results (as reported in the original papers)
Chen et al. (2025)	Prompted LLM as detector	On FND23, GPT-3.5-turbo with a task-specific prompt reports $\text{Acc}=0.875$ and $\text{Macro-}F_1=0.528$; the paper additionally benchmarks multiple datasets/prompting variants.
Raza et al. (2025)	LLMs vs. BERT-like (zero-/few-shot, fine-tune)	The authors report F_1 : on their GPT-labeled dataset vs. NELA-GT, RoBERTa achieves 0.897 (vs. 0.836), while Mistral-7B-v0.2 fine-tuned achieves 0.802 (vs. 0.737).
Choi et al. (2025)	LLM-based data augmentation	Finds LLM-generated augmentation yields limited or inconsistent gains for COVID-19 misinformation stance detection across settings, cautioning that fluency does not guarantee task-useful diversity.
This work (SharP)	Boundary-aware LLM-guided augmentation (hard + soft prompts)	PHEME (5 events, 1%–70% labeled): Acc improves from 0.760 (1%) to 0.938 (70%); consistently outperforms strong baselines under low resource splits (see Table 6).

further demonstrated that optimizing continuous prompt embeddings alone can be competitive for downstream tasks, solidifying “trainable prompts” as a practical paradigm for parameter-efficient adaptation. These techniques have been particularly attractive for few-shot and low resource settings due to their reduced training cost. Related studies also examined how prompt design interacts with few-shot classification performance in large models such as GPT-3 (Zhao et al., 2021). More recently, Zhu et al. (2024) proposed Soft Knowledgeable Prompt-Tuning (SKPT), integrating external knowledge to enhance short-text classification under limited-label regimes.

Overall, the literature has progressed from discrete prompt engineering to continuous, trainable prompting, and further toward incorporating structured priors such as external knowledge. Despite the effectiveness of prompting, directly applying it to fake news augmentation remains challenging because misinformation spans multiple domains and styles: hard prompts may lack adaptability, while soft prompts require task-aligned optimization signals to generate domain-consistent and label-preserving samples.

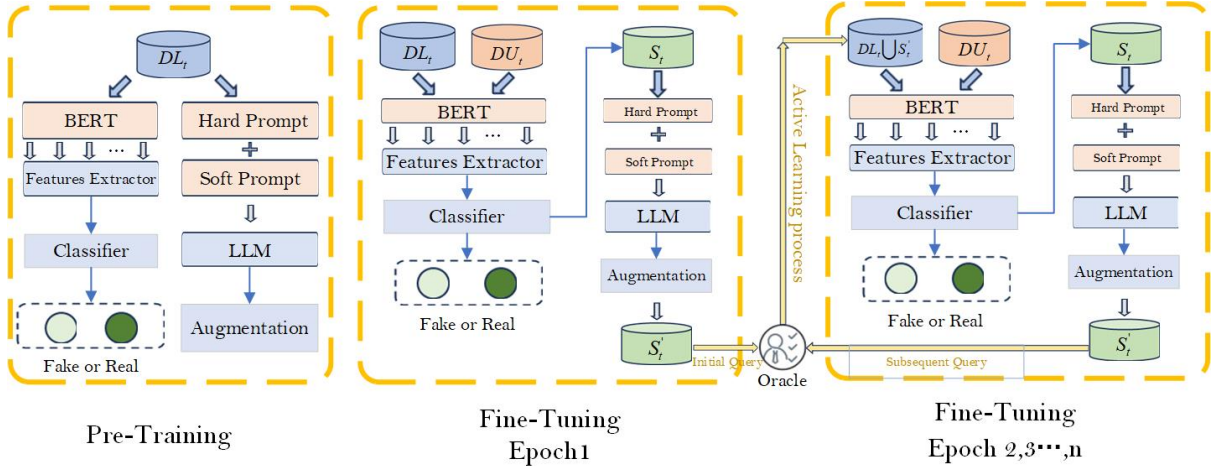


Figure 1: Model Architecture of SharP

3. Method

In this section, we first formally describe the problem, then introduce the key modules of the fake news detection model, and provide a detailed explanation of the data

Table 3: Notation used in the Method section

Symbol	Description
$DL_t = (X_L, Y_L)$	Labeled pool at round t , where X_L and Y_L denote labeled texts and their labels.
$DU_t = X_U$	Unlabeled pool at round t , where X_U denotes unlabeled texts.
S_t	Queried unlabeled batch selected from DU_t at round t .
S'_t	Augmented labeled set generated from S_t (inherits labels).
b	Query batch size in each active learning round.
T	Total number of active learning rounds.
$x_u^{(i)} / T_{\text{in}}^{(i)}$	Input text sequence of the i -th queried instance (we use $x_u^{(i)} \equiv T_{\text{in}}^{(i)}$ when convenient).
$x_u^{(i)'} / T_{\text{out}}^{(i)}$	Augmented (generated) text for the i -th queried instance (we use $x_u^{(i)'} \equiv T_{\text{out}}^{(i)}$ when convenient).
$\mathcal{H}(x)$	Predictive entropy (uncertainty) of the classifier for input x .
$P(y x)$	Predicted posterior probability of $y \in \{0, 1\}$ for input x .
τ	Entropy threshold for selecting boundary samples.
$\mathcal{B}_t$	Boundary sample set at round t selected by $\mathcal{H}(x) \geq \tau$.
T_h	Hard prompt (discrete instruction tokens).
T_s	Soft prompt vector (learnable continuous prefix).
$\oplus$	Concatenation operator.
$\odot$	Element-wise (Hadamard) product.
T_{concat}	Input concatenated with hard prompt.
E_{combined}	Combined embedding fed into the LLM (after adding T_s).
H_{out}	Last-layer hidden representation of the LLM output.
$F(\cdot)$	Feature extractor.
$D(\cdot)$	Classifier.
$F_{\text{in}}, F_{\text{out}}$	Feature vectors of the input and generated representations.
c, c^*	Classifier score and target score.
λ_1, λ_2	Weights for combining the two objectives in prompt tuning.
$\lambda_{\text{reg}}, \delta$	Regularization weight and minimum-distance margin.
η	Learning rate.
N_1, N_2, N	Epochs for training D/F , training T_s , and fine-tuning D per round.

augmentation strategy and tuning methods. The proposed model consists of two core modules: the classifier module and the data augmentation module. The classifier module integrates a feature extractor and a classifier, responsible for identifying the authenticity of fake news. The data augmentation module, centered on an LLM, combines hard prompts and soft prompts to guide the model in generating augmented text, aiming to alleviate the problem of scarce training samples and improve the model’s detection ability and generalization performance in low resource environments.

3.1. Problem Definition

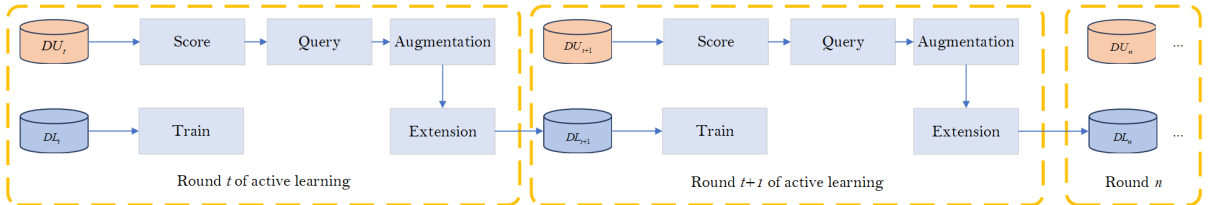


Figure 2: Method Flow

We initialize the labeled sample pool as $DL_0 = (X_L, Y_L)$ and the unlabeled sample

pool as $DU_0 = X_U$. At the t -th round of active learning ($t = 0, \dots, T - 1$), we denote the current labeled and unlabeled pools by DL_t and DU_t , respectively. A mini-batch of unlabeled instances is queried for labeling, where b is the query batch size. After obtaining labels $\{y_u^{(i)}\}_{i=1}^b$ for the queried instances, we denote the labeled queried set as:

$$S_t = \{(x_u^{(i)}, y_u^{(i)})\}_{i=1}^b. \quad (1)$$

The corresponding labels $\{y_u^{(i)}\}_{i=1}^b$ are obtained (e.g., from human annotators or an auxiliary labeling model), and the classifier is subsequently retrained on the enlarged labeled set. For each queried instance $x_u^{(i)}$, the augmentation module generates a synthetic counterpart $x_u^{(i)'}$, resulting in the augmented set:

$$S'_t = \{(x_u^{(i)'}, y_u^{(i)})\}_{i=1}^b \quad (2)$$

where each synthetic sample inherits the label of its source instance, i.e., $(x_u^{(i)'}, y_u^{(i)})$. Finally, the labeled and unlabeled pools are updated as follows:

$$\begin{aligned} DL_{t+1} &= DL_t \cup S_t \cup S'_t, \\ DU_{t+1} &= DU_t \setminus \{x \mid (x, y) \in S_t\}. \end{aligned} \quad (3)$$

3.2. Active Learning Module

We design a classifier model integrated with a feature extractor and combine it with an active learning strategy to select samples near the model’s decision boundary, i.e., those for which the model exhibits the highest uncertainty in its predictions. These samples are highly valuable for model training and will be used for data augmentation in subsequent steps, further enhancing model performance.

To effectively extract emotional features, we construct emotion representations based on five categories of emotional lexicons: Emotion, Sentiment, Morality, Imageability, and Hyperbole. We use the NRC Emotion Lexicon (Mohammad and Turney, 2013) to extract 8 emotional features, and the NRC Sentiment Lexicon and the Moral Foundations Dictionary (Graham et al., 2009) to extract emotional and moral features (2 and 10, respectively). Additionally, we leverage the MRC Psycholinguistic Database (Wilson, 1988) to extract 2 Imageability-related features and construct emotional features by analyzing approximately 350 hyperbole words.

Finally, these multi-dimensional emotional features are integrated into a unified emotional vector, representing the diverse emotional information in the news.

We design a dual feature extraction model that combines sentiment and textual features based on feature extraction. The model comprises two key components: the first component uses Convolutional Neural Networks to extract word-level latent representations of the text (v_{word}). Specifically, the input text sequence is encoded by BERT as $S = \{w_1, w_2, \dots, w_n\}$, where w_i is the embedding of the i -th word. The CNN applies multi-scale convolutions to the word embeddings, using kernel sizes of 3, 4, and 5 to extract vertical features. The max pooling layer selects the maximum value from each feature map generated by the convolution operations, forming a pooled feature vector and producing the text’s latent representation (v_{word}):

$$v_{\text{word}} = \text{MaxPool}(\text{CNN}(w_i)). \quad (4)$$

The second component uses a Bidirectional Gated Recurrent Unit to capture sentiment flow in the text, producing a sentiment representation (v_{affect}). Here, e represents the sentiment vector:

$$v_{\text{affect}} = \text{Bi-GRU}(e). \quad (5)$$

After obtaining the word-level representation (v_{word}) and sentiment features (v_{affect}), we combine them using element-wise multiplication to generate a unified text-sentiment representation (v_{fuse}). This step deeply fuses sentiment and text information, enhancing the model’s ability to perceive sentiment within the text:

$$v_{\text{fuse}} = v_{\text{word}} \odot v_{\text{affect}}. \quad (6)$$

The classifier, which utilizes the features extracted by the feature extractor, distinguishes between real and fake samples. It is a multilayer perceptron consisting of a fully connected layer and a softmax layer.

During the pre-training phase, the feature extractor and classifier are jointly trained to learn initial feature representations and classification abilities. In the fine-tuning phase, an active learning strategy is applied to guide the sample selection process.

Based on the classifier’s current predictions, we select samples near the decision boundary, i.e., those with the highest predictive uncertainty. In classification tasks, entropy is adopted as the uncertainty measure for the model’s prediction on each sample. Sun et al. (2025)

For a binary classification task, given a sample x , the classifier $D(x)$ outputs the posterior probability of belonging to class 1 as $P(y = 1 | x)$, and to class 0 as $P(y = 0 | x) = 1 - P(y = 1 | x)$. The predictive entropy $\mathcal{H}(x)$ is computed as:

$$\mathcal{H}(x) = -P(y = 1 | x) \log(P(y = 1 | x)) - P(y = 0 | x) \log(P(y = 0 | x)). \quad (7)$$

Using a threshold τ , we define the boundary sample set as:

$$\mathcal{B}_t = \{x \in DU_t \mid \mathcal{H}(x) \geq \tau\}. \quad (8)$$

These samples are highly valuable for enhancing model performance. The selected samples are annotated either manually or via automated labeling tools to ensure label accuracy. Once labeled, the samples are forwarded to the data augmentation module, where augmented texts are generated via a hybrid strategy of hard and soft prompts. These augmented samples retain the semantics and labels of the original inputs, thereby increasing the diversity of the training set. The augmented and original labeled samples are jointly used to fine-tune the model in the current training round.

This process is performed iteratively: after each round of fine-tuning, the active learning module selects a new batch of samples for augmentation and training, thereby progressively improving model performance.

3.3. Data Augmentation Module

The core of the data augmentation module lies in the training of soft prompts. By learning soft prompts, implicit knowledge from the fake news domain, such as high-frequency argumentation structures, is injected into the Transformer attention mechanism. This effectively guides the large model to perform data augmentation on news

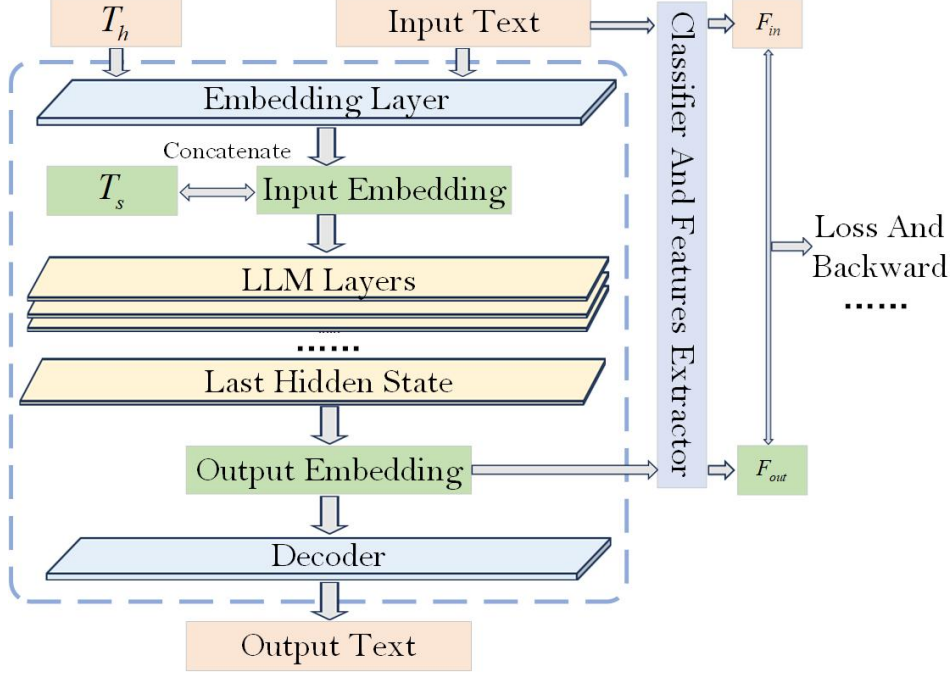


Figure 3: Architecture of Data Augmentation Module

samples from specific domains, thereby enhancing the domain adaptability of the large model and achieving fine-grained generation control.

Specifically, we first define a list of hard prompts containing various instructions to guide the large model in generating texts with different styles. These hard prompts are combined with the input text, and after passing through the embedding layer of the large model, we obtain the embedded representation of the input text. Next, we initialize a soft prompt vector and concatenate it with the input text’s embedding vector, using the concatenated result as input to the large model. The soft prompt vector, serving as prefix information of the input text, after training, will contain guiding information specific to the task context within the large model.

$$T_{\text{concat}} = T_{\text{in}} \oplus T_h, \quad (9)$$

$$E_{\text{combined}} = \text{Embed}(T_{\text{concat}}) \oplus T_s. \quad (10)$$

Since the model needs to adapt to news samples from different domains, training soft prompt vectors to effectively guide the large model in performing data augmentation tasks on specific datasets is crucial. Experimental results validate the effectiveness of this method, demonstrating that soft prompt vectors can significantly enhance the model’s adaptability and generalization across multiple datasets, thereby improving performance in cross-domain fake news detection tasks.

To avoid the gradient truncation issue that may arise from the softmax and sampling operations during the decoding stage of the large model, we use the vector from the last hidden layer output by the large model H_{out} as the embedded representation of the output text. This representation, along with the input text embedding, is fed into the feature extractor to obtain the respective feature vectors. Here, F denotes the feature extractor.

$$H_{\text{out}} = \text{Hidden}(E_{\text{combined}}), \quad (11)$$

$$F_{\text{out}} = F(H_{\text{out}}), \quad F_{\text{in}} = F(\text{Embed}(T_{\text{in}})). \quad (12)$$

It is important to note that although our feature extractor is based on BERT encoding, the large models used for data augmentation are typically those that perform better in text generation (such as GPT, Llama, and in our case, zephyr-7b-beta) (Tunstall et al., 2023). Therefore, directly using text embeddings generated by these large models is not suitable for the BERT-based feature extractor.

To address this issue, we first extract the vocabulary of the large model used for data augmentation and establish a mapping relationship with the BERT embedding vocabulary. Through this mapping, we can convert the text embeddings generated by the large model into BERT-compatible encodings, allowing the generated text to be directly fed into the feature extractor to obtain the text’s feature vector.

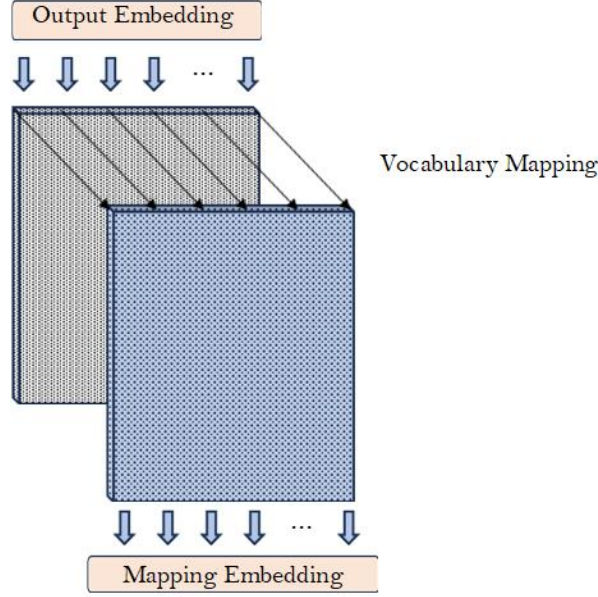


Figure 4: Vocabulary Mapping

It is worth noting that the input texts entering the data augmentation module are all located near the decision boundary of the classifier, representing the samples that are most challenging for the classifier to distinguish. The closer the generated augmented samples are to the original input text in the feature space, the more difficult it is for the classifier to classify them, thereby effectively enhancing the model’s ability to identify instances near the decision boundary.

Moreover, we explicitly require the large model, through hard prompts, to maintain the main content of the news event unchanged during the generation process, ensuring semantic consistency. This allows the generated augmented text to inherit the label of the input news sample. This design not only avoids the need for additional manual labeling of generated samples but also ensures that the augmented samples extend the decision boundary while maintaining label consistency, thereby further enhancing the model’s generalization ability.

Based on this, we first calculate the cosine similarity between the input text and the generated text in the feature space and use it as the first loss objective optimization function. This loss function ensures that the generated augmented samples maintain a high degree of semantic consistency with the input text, while simultaneously enhancing

the classifier’s discrimination ability near the decision boundary, thereby improving the overall performance of the fake news detection model.

$$\text{Sim}_{\cos}(F_{\text{in}}, F_{\text{out}}) = \frac{F_{\text{in}} \cdot F_{\text{out}}}{\|F_{\text{in}}\| \|F_{\text{out}}\|}. \quad (13)$$

To avoid generating trivial samples (i.e., those that entirely overlap with the input in the feature space), we introduce a regularization term. When the Euclidean distance between the input and generated sample falls below a predefined threshold, a penalty is applied, effectively enforcing a minimum distance between them. This promotes diversity in the generated samples and improves the overall quality of augmentation.

Here, λ_{reg} denotes the regularization weight, δ is the minimum acceptable distance, and $D_{\text{euclid}}(F_{\text{in}}, F_{\text{out}})$ represents the Euclidean distance:

$$\text{Loss}_1 = -\text{Sim}_{\cos}(F_{\text{in}}, F_{\text{out}}) + \text{Reg}, \quad (14)$$

$$\text{Reg} = \lambda_{\text{reg}} \cdot \max(0, \delta - D_{\text{euclid}}(F_{\text{in}}, F_{\text{out}})). \quad (15)$$

To achieve end-to-end joint training and the collaborative goal of semantic fidelity and decision optimization, we further introduce a second loss function, integrating the training of the data augmentation module and the classifier module. Specifically, the generated text is evaluated by the classifier, and the loss returned by the classifier serves as the second loss function.

$$c = D(H_{\text{out}}), \quad \text{Loss}_2 = (c - c^*)^2. \quad (16)$$

Through this design, we ensure that the generated news samples exhibit high uncertainty for the classifier, thereby enhancing the model’s ability to learn from hard-to-classify samples. Finally, we perform a weighted summation of the first objective loss (feature space similarity loss) and the second objective loss (classifier loss) to form the total objective loss. We then optimize the soft prompt vectors at the input end of the large model through backpropagation. This optimization process not only improves the generation quality of the data augmentation module but also achieves collaborative optimization with the classifier module, thereby further enhancing the overall performance of the model in fake news detection tasks.

$$\text{Loss}_{\text{total}} = \lambda_1 \cdot \text{Loss}_1 + \lambda_2 \cdot \text{Loss}_2, \quad (17)$$

$$T_s \leftarrow T_s - \eta \nabla_{T_s} \text{Loss}_{\text{total}}. \quad (18)$$

3.4. Algorithm Optimization

The training pipeline consists of two major stages: pre-training and fine-tuning. The model architecture comprises two key modules: a classifier and a data augmentation module. To maximize detection performance, we separately optimize the classifier and the augmentation module during each stage.

Pre-training Stage: In the pre-training stage, we begin by training the classifier. The feature extractor encodes both textual and emotional information into feature representations, which serve as inputs to the classifier. The classifier is trained to minimize detection loss, enabling the model to distinguish fake news from real news based on the

initial labeled dataset and laying a foundation for subsequent fine-tuning and augmentation.

Following this, the data augmentation module is trained. The goal of this stage is to initialize and optimize the soft prompt vectors, enabling the language model to generate high-quality augmented texts conditioned on the input. Through a dual objective optimization mechanism, the soft prompts enforce semantic consistency between the generated and original texts, while also increasing the sample density near the decision boundary to support the classifier in later stages.

Fine-tuning Stage:

During the fine-tuning phase, we adopt an active learning loop to progressively expand supervision and refine both the prompt-based augmentation module and the classifier. At round t , the current classifier first scores unlabeled instances in DU_t and selects a batch S_t near its decision boundary for annotation. After obtaining labels for S_t , we first fine-tune the soft prompt vector T_s on these newly labeled samples, while freezing all parameters of the LLM backbone. This prompt-tuning step adapts T_s to the current data distribution and makes the generation process more task- and domain-aligned.

Next, the augmentation module uses the *fine-tuned* T_s (together with hard prompts T_h) to generate an augmented set S'_t from S_t , inheriting labels from their corresponding inputs. The labeled pool is then updated by merging the newly labeled samples and their augmentations. Finally, we fine-tune the classifier by standard supervised training on the updated labeled pool, so that it can better separate hard instances around the boundary. Repeating this procedure for multiple rounds yields a jointly refined augmentation module (via prompt tuning) and classifier (via supervised fine-tuning under the expanded labeled pool), enabling robust detection in label-scarce scenarios.

Algorithm 1: Model Pre-training Process

Input: Initial labeled pool $DL_0 = (X_L, Y_L)$, Hard prompt T_h , Feature extractor $F(\cdot)$, Classifier $D(\cdot)$; Learning rate η , Number of epochs N_1 and N_2

Output: Trained soft prompt vector T_s , Updated model parameters of F and D

Step 1: Train classifier and feature extractor on DL_0

for $epoch = 1$ to N_1 **do**

 Sample $(x_L, y_L) \sim DL_0$;

$$\mathcal{L}_D = \text{Loss}_D(D(x_L), y_L)$$

$$\mathcal{L}_F = \text{Loss}_F(F(x_L), y_L)$$

$$D \leftarrow D - \eta \nabla_D \mathcal{L}_D, \quad F \leftarrow F - \eta \nabla_F \mathcal{L}_F$$

Step 2: Train soft prompt T_s on DL_0

Initialize soft prompt vector T_s ;

for $epoch = 1$ to N_2 **do**

 Sample $(x_L, y_L) \sim DL_0$;

$$T_{\text{concat}} = x_L \oplus T_h$$

$$E_{\text{combined}} = \text{Embed}(T_{\text{concat}}) \oplus T_s$$

$$H_{\text{out}} = \text{Hidden}(E_{\text{combined}})$$

$$F_{\text{out}} = F(H_{\text{out}}), \quad F_{\text{in}} = F(\text{Embed}(x_L))$$

$$\text{Sim}_{\cos}(F_{\text{in}}, F_{\text{out}}) = \frac{F_{\text{in}} \cdot F_{\text{out}}}{\|F_{\text{in}}\| \|F_{\text{out}}\|}$$

$$\text{Loss}_1 = -\text{Sim}_{\cos}(F_{\text{in}}, F_{\text{out}}) + \text{Reg}$$

$$T_s \leftarrow T_s - \eta \nabla_{T_s} \text{Loss}_1$$

return Trained soft prompt vector T_s , Updated model parameters of F and D

Algorithm 2: Model Fine-tuning Process

Input: Initial labeled pool $DL_0 = (X_L, Y_L)$, Initial unlabeled pool $DU_0 = X_U$, Trained soft prompt vector T_s , Hard prompt T_h , Feature extractor $F(\cdot)$, Classifier $D(\cdot)$; Learning rate η , Number of active learning rounds T , Query batch size b ; Number of epochs N_2 and N per round; Weights λ_1, λ_2 , Target score c^*

Output: Fine-tuned model parameters, Final labeled pool DL_T

Initialize $DL_0 = (X_L, Y_L)$, $DU_0 = X_U$;

for $t = 0$ **to** $T - 1$ **do**

Step 1: Active learning and labeling

 Select and query b instances from DU_t according to the active learning strategy, and obtain labels to form

$$S_t = \{(x_u^{(i)}, y_u^{(i)})\}_{i=1}^b.$$

$$DU_{t+1} = DU_t \setminus \{x \mid (x, y) \in S_t\}.$$

Step 2: Fine-tune soft prompt T_s (loss differs from pre-training)

 Update T_s by repeating the soft prompt training procedure in Step 2 of Algorithm 1 on the batch S_t , except that the objective is replaced by

$$\text{Loss}_{\text{total}} = \lambda_1 \cdot \text{Loss}_1 + \lambda_2 \cdot \text{Loss}_2, \quad \text{Loss}_2 = (c - c^*)^2, \quad c = D(H_{\text{out}}),$$

 while all other computations (including T_{concat} , E_{combined} , H_{out} , F_{in} , F_{out} and Loss_1) follow Algorithm 1;

$$T_s \leftarrow T_s - \eta \nabla_{T_s} \text{Loss}_{\text{total}}$$

Step 3: Generate augmented set S'_t from S_t using the updated T_s

 Initialize $S'_t = \emptyset$;

for $i = 1$ **to** b **do**

$$T_{\text{concat}}^{(i)} = x_u^{(i)} \oplus T_h$$

$$E_{\text{combined}}^{(i)} = \text{Embed}(T_{\text{concat}}^{(i)}) \oplus T_s$$

$$H_{\text{out}}^{(i)} = \text{Hidden}(E_{\text{combined}}^{(i)})$$

$$T_{\text{out}}^{(i)} = \text{Decode}(H_{\text{out}}^{(i)})$$

 Add $(T_{\text{out}}^{(i)}, y_u^{(i)})$ to S'_t ;

$$S'_t = \{(T_{\text{out}}^{(i)}, y_u^{(i)})\}_{i=1}^b$$

Step 4: Update labeled pool and train classifier

$$DL_{t+1} = DL_t \cup S_t \cup S'_t$$

for $\text{epoch} = 1$ **to** N **do**

 Sample $(x, y) \sim DL_{t+1}$;

$$\mathcal{L}_D = \text{Loss}_D(D(x), y)$$

$$D \leftarrow D - \eta \nabla_D \mathcal{L}_D$$

return Fine-tuned model parameters, Final labeled pool DL_T

4. Experiments

In the experimental section, we aim to address the following key questions:

- Does the SharP model demonstrate significant performance advantages in low resource environments while maintaining satisfactory performance under varying resource conditions?
- Does the model exhibit good generalization ability across news data from different domains?
- Can the texts generated by the data augmentation module effectively enhance the classifier’s detection performance?
- Do different hard prompts and other hyperparameter choices have a significant impact on the model’s performance?

We systematically evaluate SharP by first describing the experimental setup and then conducting comprehensive tests across various conditions to validate its effectiveness in fake news detection.

4.1. Experimental Setup

Datasets: In Group A, we evaluate our model on five distinct event-specific datasets: *Germanwings-crash*, *Sydneysiege*, *Ottawashooting*, *Charliehebd*, and *Ferguson*. These subsets are derived from the PHEME dataset (Zubiaga et al., 2016), which consists of multimodal Twitter data, including users’ social information, posted content, and annotated labels. The PHEME dataset is particularly well-suited for fake news detection tasks, as it not only offers rich textual content but also provides sentiment annotations and propagation path information for each tweet. In our experimental design, we proportionally divide the training samples within each subset to comprehensively test model performance under varying sample sizes and across diverse news scenarios.

In addition to the event-specific evaluation on PHEME, we further conduct a cross-benchmark study (Group B) on two widely used misinformation benchmarks, LIAR (Wang, 2017) and Twitter15/16 (Ma et al., 2017a,b), to assess the robustness of SharP across datasets with different data characteristics. LIAR consists of short, fact-checked political claims annotated with fine-grained veracity labels, which are commonly consolidated into binary labels for fake news detection. Twitter15/16 are event-centric rumor detection benchmarks built from Twitter conversations with reply structures, providing complementary evaluation beyond the PHEME breaking-news threads.

Dataset	Cha.	Fer.	Ger.	Ott.	Syd.	Total
Labeled Fake News	318	144	95	330	382	1269
Labeled Real News	1481	719	94	280	559	3133
Unlabeled Fake News	140	140	143	140	140	703
Unlabeled Real News	140	140	137	140	140	697

Table 4: Dataset statistics for Group A (PHEME) used in our labeled/unlabeled pool setting

Dataset	LIAR	Twitter15/16
Labeled Fake News	5,514	445
Labeled Real News	6,997	429
Unlabeled Fake News	143	134
Unlabeled Real News	137	146
Total	12,791	1,154

Table 5: Dataset statistics for Group B (LIAR and Twitter15/16) used in our labeled/unlabeled pool setting

Experimental Details: To comprehensively evaluate the model’s performance under varying resource conditions, we divide the experimental dataset into three subsets: the first group is the labeled data pool used for pre-training, the second group is a fixed-size unlabeled data pool, and the third group is the test data pool.

During the pre-training phase, the model is trained on the labeled data pool to acquire fundamental text classification capabilities. Subsequently, the model proceeds to the fine-tuning phase, where it undergoes further optimization using the unlabeled data pool. In this phase, an active learning strategy is employed to select samples near the decision boundary for labeling and training, thereby enhancing the model’s performance through targeted sample selection.

In the experiment, we operationalize *low resource* fake news detection by constraining the size of the labeled training pool. Concretely, for each dataset, we construct a series of label budgets by randomly sampling $\{1\%, 5\%, 10\%, 15\%, 20\%, 25\%, 30\%, 35\%, 40\%, 45\%, 50\%, 70\%\}$ of instances from the labeled pool for training. To reduce sampling bias under small budgets, we adopt stratified sampling to preserve the fake/real ratio (and keep the split as balanced as possible). We then evaluate performance across these budget levels to assess robustness under varying degrees of supervision scarcity, with particular attention to whether the proposed augmentation strategy yields consistent gains when labeled data are limited. Unless otherwise specified, all other settings are kept unchanged across budget levels.

4.2. Baseline

In the baseline experiments, we select a series of classical and advanced fake news detection methods as baseline models. We evaluate and compare their performance with the proposed method under the same datasets and experimental settings, with a primary focus on assessing the model robustness and generalization ability in low resource environments.

Here are the baseline models we selected:

HAN (2016): A fake news detection method based on the Hierarchical Attention Network (HAN). This method utilizes a hierarchical attention mechanism to model different levels of information in news text (such as words, sentences, and paragraphs), thereby capturing essential semantic features. HAN effectively focuses on critical information within the text and combines structural features of long texts, enhancing the model’s classification performance in fake news detection tasks (Yang et al., 2016).

dEFEND (2019): A multimodal fake news detection model that integrates textual content and user comments. dEFEND uses explainable attention mechanisms to identify

suspicious words and responses, enhancing model interpretability and robustness (Shu et al., 2019).

HMCAN (2021): A fake news detection method based on a hierarchical multimodal contextual attention network, which integrates text and social network comment information. It leverages CNNs and hierarchical attention to extract features from multimodal data, aiming to improve classification accuracy (Qian et al., 2021).

DDT (2025): A fake news detection method based on active learning, which enhances the model’s discrimination ability by combining textual features and emotional features. In addition, DDT employs an active learning strategy, dynamically selecting uncertain samples near the decision boundary for manual annotation and fine-tuning during training. This approach maximizes the model’s learning efficiency under limited labeled data conditions. (Hu et al., 2025)

DeepSeek (2024): A general-purpose language model based on the Transformer architecture, incorporating various structural optimization techniques, such as RMSNorm, SwiGLU activation function, and RoPE (Rotary Positional Embedding), to improve performance in long-text modeling and reasoning efficiency. The model is pre-trained on large-scale bilingual (Chinese and English) corpora, demonstrating strong cross-linguistic understanding and generalization capabilities. In this study’s baseline experiments, we fine-tune DeepSeek-7B-Base using LoRA (Low-Rank Adaptation) on data with different resource proportions to evaluate its performance and adaptability in the fake news detection task (Liu et al., 2024).

4.2.1. Group A: Event-specific evaluation on PHEME

To comprehensively simulate real-world application scenarios with varying data resources, we construct multiple experimental groups with different training sample sizes based on five sub-datasets. The specific setup is as follows: we randomly sample from the original dataset in proportional ratios to build five resource scenarios: few-shot (1%, 5%), low resource (10%, 20%), mid resource (30%, 40%), high resource (50%), and near-full (70%), resulting in a total of eight grouped scenarios. Each group is independently trained and tested on the five domain-specific sub-datasets to thoroughly assess the performance and generalization ability of each method under conditions of varying sample sizes and domain distribution.

This experimental setup not only reflects the model’s adaptability under extreme few-shot conditions but also allows for measuring the performance trend as data resources increase. This provides empirical evidence for data augmentation strategies in future low resource text classification tasks.

Dataset	Stra.	1%	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	70%
Germanwings-crash	HAN	0.500	0.734	0.713	0.691	0.691	0.745	0.734	0.755	0.723	0.766	0.745	0.713
	dEFEND	0.670	0.745	0.723	0.702	0.745	0.755	0.745	0.787	0.681	0.745	0.755	0.713
	HMCAN	0.250	0.272	0.652	0.757	0.766	0.712	0.610	0.727	0.651	0.750	0.768	0.813
	DDT	0.740	0.729	0.760	0.781	0.792	0.823	0.833	0.854	0.865	0.854	0.885	0.927
	Deepseek-7b	0.617	0.617	0.617	0.617	0.628	0.628	0.628	0.617	0.606	0.617	0.617	0.606
	SharP	0.760	0.802	0.792	0.833	0.833	0.813	0.854	0.885	0.875	0.875	0.917	0.938
Sydneyseige	HAN	0.500	0.638	0.681	0.638	0.670	0.713	0.691	0.723	0.660	0.670	0.702	0.745
	dEFEND	0.606	0.660	0.628	0.617	0.681	0.691	0.660	0.691	0.702	0.691	0.702	0.681
	HMCAN	0.250	0.272	0.630	0.714	0.734	0.695	0.667	0.749	0.651	0.750	0.768	0.813
	DDT	0.677	0.708	0.771	0.771	0.760	0.771	0.823	0.865	0.906	0.885	0.885	0.906
	Deepseek-7b	0.462	0.441	0.462	0.462	0.462	0.462	0.462	0.462	0.462	0.462	0.462	0.462
	SharP	0.760	0.750	0.823	0.802	0.833	0.844	0.833	0.875	0.906	0.927	0.906	0.906
Ottawashooting	HAN	0.500	0.649	0.681	0.723	0.638	0.670	0.713	0.606	0.638	0.681	0.691	0.660
	dEFEND	0.670	0.638	0.691	0.660	0.681	0.649	0.691	0.596	0.617	0.660	0.670	0.660
	HMCAN	0.250	0.272	0.630	0.714	0.766	0.737	0.667	0.727	0.651	0.727	0.750	0.813
	DDT	0.583	0.688	0.646	0.698	0.740	0.792	0.802	0.833	0.771	0.896	0.885	0.906
	Deepseek-7b	0.500	0.500	0.500	0.500	0.489	0.489	0.500	0.489	0.489	0.489	0.489	0.489
	SharP	0.695	0.719	0.750	0.813	0.740	0.823	0.854	0.844	0.854	0.917	0.896	0.927
Charliehebd	HAN	0.500	0.702	0.734	0.691	0.734	0.702	0.713	0.691	0.734	0.723	0.755	0.755
	dEFEND	0.691	0.691	0.702	0.638	0.723	0.702	0.755	0.638	0.734	0.702	0.691	0.766
	HMCAN	0.250	0.272	0.735	0.757	0.766	0.737	0.631	0.727	0.651	0.764	0.768	0.817
	DDT	0.708	0.750	0.792	0.792	0.823	0.833	0.844	0.885	0.865	0.865	0.896	0.885
	Deepseek-7b	0.362	0.362	0.372	0.372	0.372	0.372	0.372	0.372	0.383	0.372	0.383	0.383
	SharP	0.813	0.792	0.823	0.813	0.833	0.844	0.844	0.854	0.906	0.865	0.927	0.917
Ferguson	HAN	0.500	0.755	0.723	0.713	0.681	0.691	0.755	0.766	0.734	0.809	0.787	0.723
	dEFEND	0.638	0.723	0.702	0.691	0.723	0.734	0.713	0.755	0.691	0.723	0.734	0.702
	HMCAN	0.250	0.272	0.735	0.757	0.745	0.737	0.610	0.727	0.651	0.764	0.768	0.817
	DDT	0.708	0.729	0.750	0.781	0.813	0.813	0.771	0.854	0.844	0.865	0.875	0.927
	Deepseek-7b	0.681	0.681	0.692	0.681	0.692	0.692	0.692	0.702	0.692	0.702	0.692	0.681
	SharP	0.792	0.813	0.813	0.833	0.844	0.844	0.865	0.875	0.875	0.883	0.918	0.948

Table 6: Performance comparison on Group A datasets (Acc.)

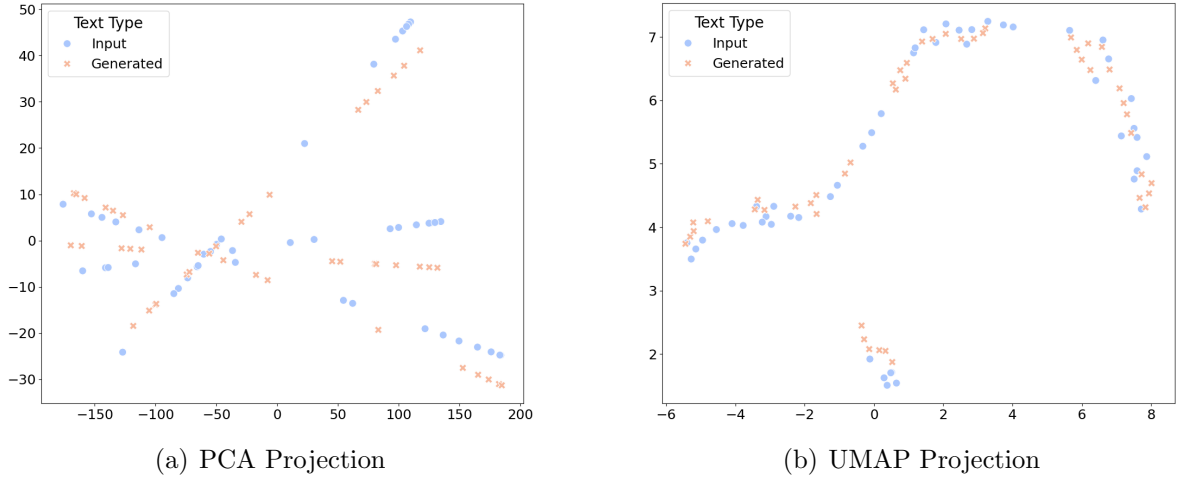


Figure 5: Sample Space Visualization

Figure 5 presents the visualization results of the sample space, intuitively showing the distribution relationship between input samples and generated samples in the feature space. The visualization results demonstrate that the trained model is capable of generating enhanced texts that closely align with input samples within the feature space. This

verifies the effectiveness of the dual prompt mechanism (hard prompt and soft prompt) and the dual objective optimization strategy (fidelity and discriminability). Furthermore, the results indicate that the generated samples are effectively guided towards the vicinity of the classifier’s decision boundary, enhancing the model’s ability to learn from hard-to-classify samples, thereby providing the most valuable augmented data for training.

Table 6 displays the experimental results of various baseline models. Compared to existing fake news detection methods, the proposed model exhibits significant overall performance improvements, particularly in experimental settings with low sample ratios. The introduction of the data augmentation module effectively improves detection accuracy, fully validating the effectiveness and practicality of the proposed method in low resource environments.

As the number of training samples increases, although the performance gap between models narrows, the proposed method consistently maintains a leading position across multiple datasets and various sample configurations. This result indicates that the proposed augmentation mechanism significantly alleviates the problem of insufficient labeled samples, enhancing the model’s robustness and detection capability in data-scarce scenarios.

Moreover, even in cases with sufficient samples, the proposed method still exhibits superior detection performance compared to other baseline models, further demonstrating its universality and advantage in the fake news detection task.

4.2.2. Group B: Cross-benchmark evaluation

As a complement to the Group A evaluation, we further assess SharP on LIAR and Twitter15/16 to examine its cross-benchmark generalization and discrimination capability across different misinformation domains and instance formats. For LIAR, we collapse the original six-way veracity labels—*pants-fire*, *false*, *barely-true*, *half-true*, *mostly-true*, and *true*—into a binary setting (*Fake* vs. *Real*) by mapping $\{\textit{pants-fire}, \textit{false}, \textit{barely-true}\}$ to *Fake* and $\{\textit{half-true}, \textit{mostly-true}, \textit{true}\}$ to *Real*. For Twitter15/16, we adopt the commonly used binary rumor verification setting (rumor vs. non-rumor) and map it to *Fake/Real*, reporting results on the merged Twitter15/16 benchmark. We follow the same low resource protocol as in Group A, including identical label budgets (1%–70%) with stratified sampling and the same training configuration unless stated otherwise.

Dataset	Stra.	1%	5%	10%	15%	20%	25%	30%	35%	40%	45%	50%	70%
PHEME (Germanwings-crash)	HAN	0.500	0.734	0.713	0.691	0.691	0.745	0.734	0.755	0.723	0.766	0.745	0.713
	dEFEND	0.670	0.745	0.723	0.702	0.745	0.755	0.745	0.787	0.681	0.745	0.755	0.713
	HMCAN	0.250	0.272	0.652	0.757	0.766	0.712	0.610	0.727	0.651	0.750	0.768	0.813
	DDT	0.740	0.729	0.760	0.781	0.792	0.823	0.833	0.854	0.865	0.854	0.885	0.927
	Deepseek-7b	0.617	0.617	0.617	0.617	0.628	0.628	0.628	0.617	0.606	0.617	0.617	0.606
	SharP	0.760	0.802	0.792	0.833	0.833	0.813	0.854	0.885	0.875	0.875	0.917	0.938
LIAR	HAN	0.318	0.436	0.564	0.564	0.439	0.564	0.406	0.564	0.482	0.557	0.466	0.564
	dEFEND	0.564	0.564	0.564	0.436	0.564	0.549	0.549	0.549	0.564	0.564	0.549	0.564
	HMCAN	0.185	0.185	0.185	0.259	0.431	0.431	0.259	0.259	0.185	0.185	0.259	0.431
	DDT	0.526	0.573	0.641	0.630	0.651	0.688	0.714	0.719	0.766	0.724	0.781	0.859
	Deepseek-7b	0.583	0.582	0.585	0.585	0.588	0.586	0.584	0.587	0.585	0.584	0.578	0.565
	SharP	0.609	0.615	0.662	0.688	0.688	0.703	0.734	0.734	0.750	0.740	0.807	0.880
Twitter15/16	HAN	0.481	0.589	0.654	0.706	0.710	0.645	0.701	0.736	0.654	0.823	0.861	0.628
	dEFEND	0.502	0.498	0.632	0.727	0.576	0.706	0.498	0.727	0.498	0.714	0.805	0.723
	HMCAN	0.250	0.250	0.333	0.250	0.500	0.333	0.667	0.500	0.500	0.667	0.750	0.750
	DDT	0.687	0.748	0.786	0.809	0.765	0.791	0.826	0.896	0.861	0.861	0.817	0.870
	Deepseek-7b	0.504	0.504	0.504	0.504	0.548	0.504	0.504	0.548	0.549	0.504	0.548	0.549
	SharP	0.783	0.835	0.878	0.844	0.844	0.870	0.844	0.870	0.878	0.887	0.852	0.896

Table 7: Performance comparison on Group B datasets (Acc.)

As reported in Table 7, SharP achieves strong performance on both LIAR and Twitter15/16. Across almost all label-budget configurations, SharP attains the best results and remains a close second in the remaining case, indicating that its gains are stable rather than confined to a specific setting. These complementary experiments further demonstrate that SharP generalizes well across heterogeneous benchmarks, including fact-checked political claims from LIAR and conversational microblog threads from Twitter15/16, which differ in domains, topics, and writing styles. This supports the robustness of our approach under diverse benchmarks and input characteristics.

4.3. Ablation Study

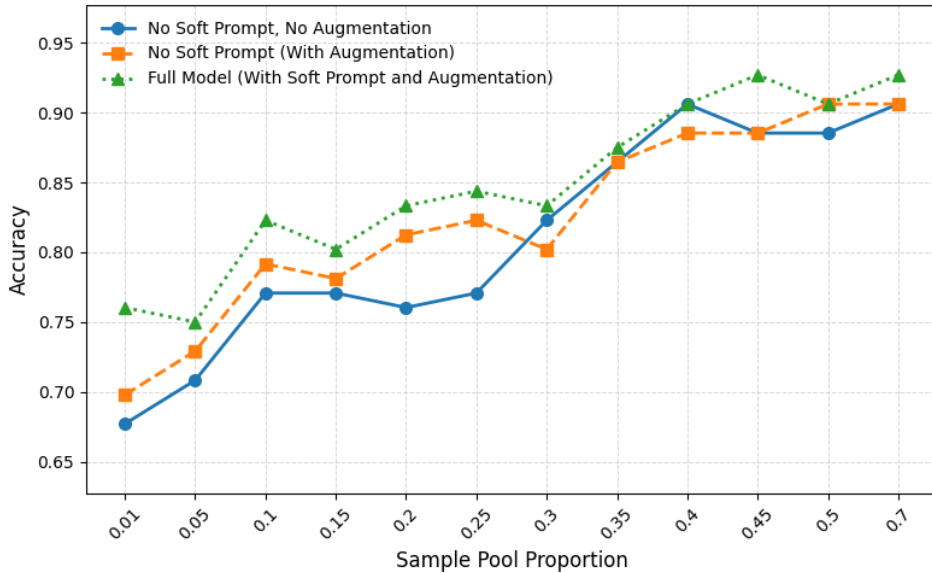


Figure 6: Ablation Study

To comprehensively evaluate the model’s performance under different component configurations, we designed an ablation study to test the following three scenarios: a single

classifier without data augmentation, a large model with data augmentation but without soft prompts, and a large model with data augmentation incorporating the dual prompt architecture. In the experiment, we conducted tests with various proportions of training samples to thoroughly assess the contribution of each component to the model’s detection performance.

The results of the ablation study further confirm the crucial role of the data augmentation module in the model. After incorporating the data augmentation module, the model’s performance improved significantly. Furthermore, when the soft prompt guidance was added, the model exhibited further improvements across all evaluation metrics, with the effects being particularly pronounced in low-sample settings. These results demonstrate that introducing large model data augmentation can significantly facilitate model training under low resource conditions. Additionally, the introduction of the dual prompt architecture enables the large model to better understand the generation requirements specific to the current dataset, leading to the creation of higher-quality augmented texts and further enhancing the overall model performance.

4.4. Hyperparameter Sensitivity Analysis

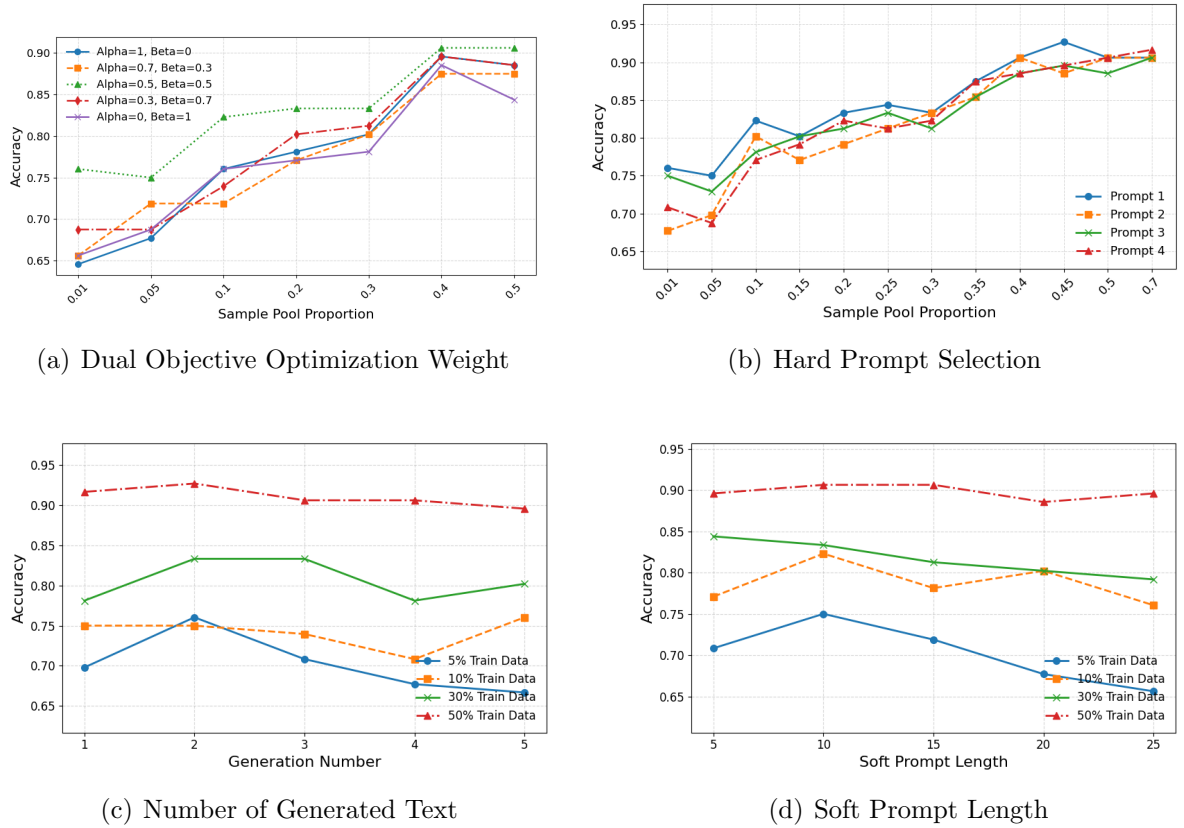


Figure 7: Hyperparameter Sensitivity Analysis

To further verify the robustness and stability of our model, we conducted a detailed hyperparameter sensitivity analysis, aiming to evaluate the impact of different hyperparameter settings on model performance. Hyperparameters play a crucial role in training deep learning models as they directly determine the convergence speed, generalization ability, and final detection performance. Therefore, selecting and adjusting hyperpa-

rameters appropriately not only enhances the model’s performance but also ensures its effectiveness across different tasks and datasets.

In this experiment, we focused on the following key hyperparameters: the choice of hard prompt categories, the size of soft prompt initialization, the number of generated data augmentation texts, and the weight settings of dual objective optimization. Adjusting each hyperparameter can influence the training dynamics of the model, thereby affecting the final detection outcomes.

By systematically analyzing these key hyperparameters, we identified which hyperparameters have the most significant impact on model performance and gained deeper insights into their functional mechanisms. This analysis provides a scientific basis and optimization direction for hyperparameter selection in practical applications. Ultimately, the hyperparameter sensitivity analysis enabled us to identify the most suitable hyperparameter combination for this task, further enhancing the model’s overall performance and generalization ability.

Figure 7 presents the experimental results of the hyperparameter analysis. In the analysis of hard prompt category selection, we observed that performance differences between different hard prompt categories were minimal when the number of training samples was large. However, when the number of training samples was small, the performance of certain hard prompt categories was significantly lower.

We hypothesize that this discrepancy may result from the classifier’s preference for specific text styles. For example, hard prompt 3 (which uses a more aggressive tone) performed better than hard prompt 4 (which uses a more declarative tone), likely because fake news often employs a more provocative and attention-grabbing style, making such texts easier to identify during classification. As the amount of training data increases, the influence of this preference on model performance gradually diminishes, leading to a reduction in performance differences.

When evaluating across all proportions of the sample pool, hard prompt 1 (which uses simpler and more straightforward expressions) consistently exhibited the best performance. Therefore, we standardized the use of hard prompt 1 in our experiments.

In the hyperparameter sensitivity analysis, we evaluated the hyperparameters under varying training data proportions (specifically 5%, 10%, 30%, and 50%) to assess their impact on model performance across different resource levels.

The experimental results indicate that as the initialization length of the soft prompt increases, the model performance gradually improves, reaching a peak before subsequently declining. This phenomenon can be attributed to the following reasons: when the soft prompt length is too short, it fails to adequately encode guiding information for the large model, resulting in lower performance. On the other hand, when the soft prompt length is too long, it occupies a substantial proportion of the input after concatenation with the model’s textual input. This excessive additional information hinders the model’s ability to effectively understand the task objective, leading to a performance decline.

In the hyperparameter analysis of the number of generated data augmentation texts, we observed that as the generation number increases, the model performance initially rises briefly, reaches a peak, and then continuously declines. Notably, when the training data volume is relatively low, the performance peak occurs with a delay.

This phenomenon can be attributed to the following reasons: when the training data is relatively sufficient, an excessive number of augmented texts can result in a surplus of semantically similar training data in the next training round, thereby inhibiting performance improvement. In contrast, when the training data volume is low, the increase

in generated text quantity enhances data diversity. In this case, the addition of more training data primarily contributes to performance improvement, resulting in a delayed arrival of the performance peak.

Finally, in the hyperparameter analysis of dual objective optimization weight settings, we found that models using only a single loss function performed worse under all three sample conditions compared to models that combined both loss functions. When appropriate weights were set and both loss functions were jointly adjusted, the model performance improved significantly. This result demonstrates the effectiveness of dual objective functions in optimizing the quality of generated samples and enhancing the overall model performance.

5. Conclusion and Future Work

In this paper, we proposed an LLM-based text data augmentation technique guided by dual prompt engineering, which combines structured hard prompts with learnable soft prompts. This approach was successfully applied to the task of fake news detection under low resource settings. Experimental results demonstrate that the proposed method significantly improves the classifier’s discriminative performance by generating diverse samples that are consistent with the domain and preserve the original labels. Particularly in low resource environments, our framework significantly enhances detection precision and overall classification accuracy by supplying diverse, high-quality augmented training samples.

Beyond empirical performance, our method provides a flexible and model independent solution that can be adapted to various LLM architectures and training setups, making it applicable to a wide range of real world misinformation detection scenarios. In addition, the sample selection strategy, which focuses on uncertain prediction regions, enables more efficient use of annotation resources in active learning settings, further demonstrating the method’s practical value.

In future work, we plan to extend this augmentation framework to a wider range of text classification tasks, including sentiment analysis, topic classification, and intent detection. We are also interested in evaluating the method’s effectiveness across multilingual datasets and under distribution shifts. In addition, we aim to explore the integration of multimodal data, such as images and user metadata, to further enhance the model’s contextual understanding and scalability in practical applications like social media content moderation and news verification systems.

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Appendix A. Hard Prompt Templates

Table A.8: Hard prompt template used for style-controlled rewriting, and an example instantiation with the *common* style (the best-performing style token in our experiments)

Item	Content
Template $T_h(x, s_t)$	Rewrite this ONE sentence using <s_t> word choices (minimal substitutions). Keep meaning/facts; keep names/numbers unchanged. Return ONE sentence only, no extra text. <x>
Style tokens	$s_t \in \{common, professional, intense, neutral\}$
Example instantiation (common style)	
Prompt ($s_t = common$)	Rewrite this ONE sentence using common word choices (minimal substitutions). Keep meaning/facts; keep names/numbers unchanged. Return ONE sentence only, no extra text. The government announced a \$5 million relief plan on Tuesday.